Deutsches Forschungszentrum für Künstliche Intelligenz
German Research Center for Artificial Intelligence

Center for European Research in Trusted Artificial Intelligence

Ministerium für Wirtschaft, Innovation, Digitales und Energie
SAARLAND

Europäischer Fonds für regionale Entwicklung im SAARLAND

Ethical Reasoning in the Dark

How to Make Justifiable Decisions When You Don't Know What's Right

Summer Institute in Computational Social Science
at the *Interdisciplinary Institute for Societal Computing* (I2SC)
SICSS-Saarbrücken 2025

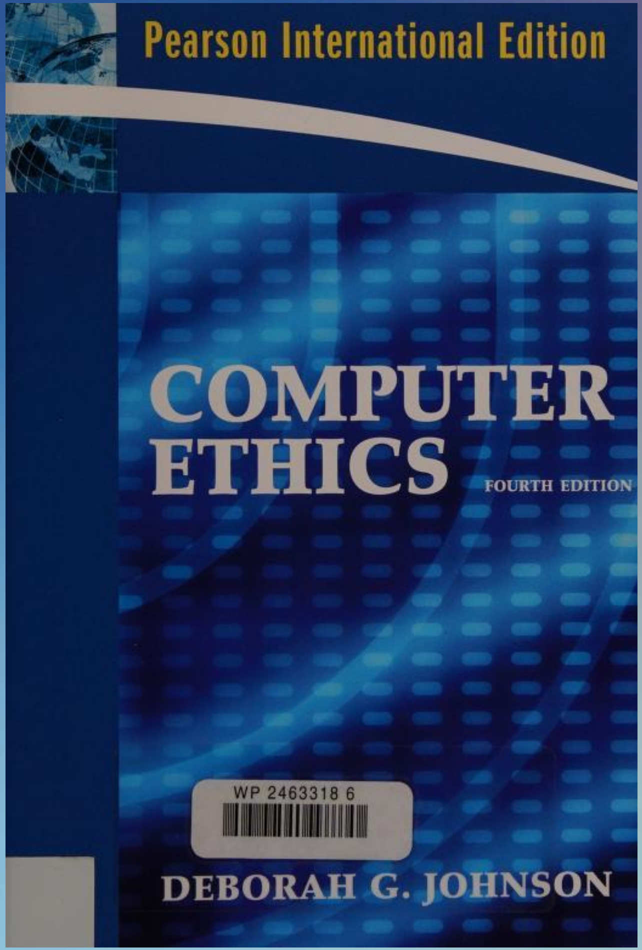
September 16, 2025

Kevin Baum (kevin.baum@dfki.de)

Co-Head of	*Center for European Research in Trusted AI* (CERTAIN)	at DFKI
Deputy of	*Neuro-Mechanistic Modeling* (NMM)	at DFKI
Lead of	*Responsible AI and Machine Ethics* (RAIME)	at NMM

Plan

- A **very short** history of **ethical challenge in context of CS**
- **Including:** What the **current topics** are and what characterizes them
- An **inconvenient truth**: the **necessity of positive normative choice** and the implication of **truly deep interdisciplinary** research
- **Practical reasoning** as a way out of **the misery of moral uncertainty**
- **Q&A**




Deborah G. Johnson


David Gotterbarn


Computer Ethics Topics

- Computers in the Workplace
- Computer Crime
- Privacy and Anonymity
- Intellectual Property
- Professional Responsibility
- Globalization
- The Meta ethics of Computer Ethics

A (very short) History of Ethics & CS

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topics many classical topics and evergreens

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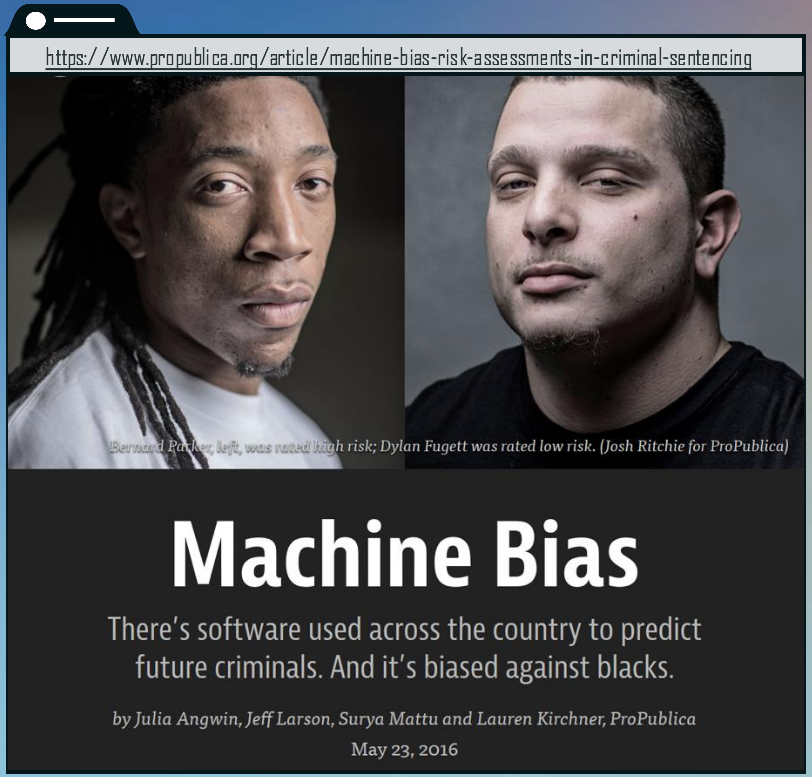
Mark Zuckerberg according to David Kirkpatrick's *The Facebook Effect* (2010)



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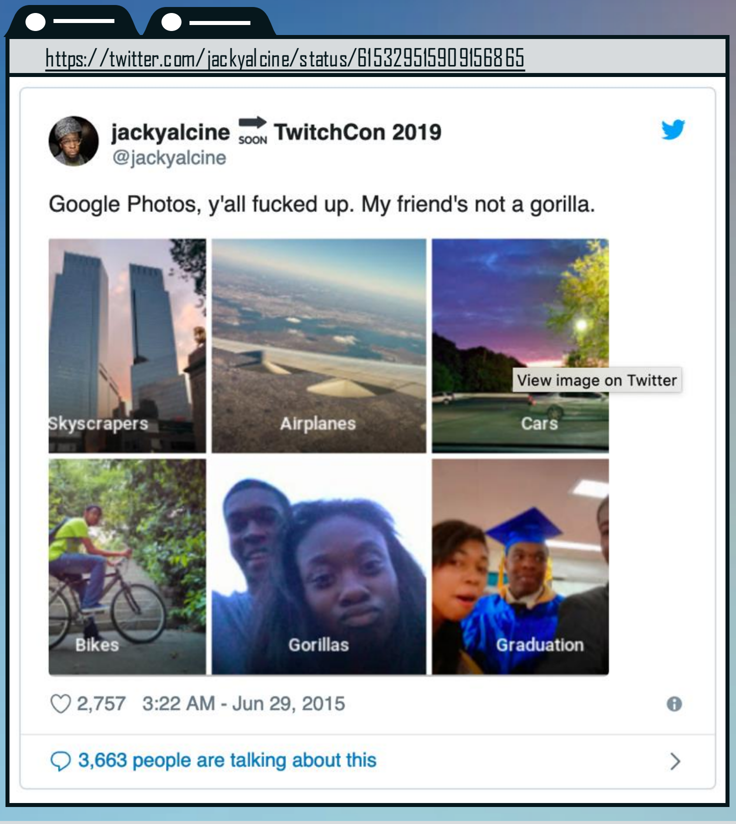

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https://vcai.mpi-inf.mpg.de/projects/MZ/Papers/CVPR2016_FF/page.html

Face2Face: Real-time Face Capture and Reenactment of RGB Videos

CVPR 2016 (Oral)

J. Thies [1] M. Zollhöfer [2] M. Stamminger [1] C. Theobalt [2] M. Nießner [3]
[1] University of Erlangen-Nuremberg [2] Max Planck Institute for Informatics [3] Stanford University

Abstract

We present a novel approach for real-time facial reenactment of a monocular target video sequence (e.g., Youtube video). The source sequenc
video stream, captured live with a commodity webcam. Our goal is to animate the facial expressions of the target video by a source ac
manipulated output video in a photo-realistic fashion. To this end, we first address the under-constrained problem of facial identity recovery f
by non-rigid model-based bundling. At run time, we track facial expressions of both source and target video using a dense photometric
Reenactment is then achieved by fast and efficient deformation transfer between source and target. The mouth interior that best matches the re-
retrieved from the target sequence and warped to produce an accurate fit. Finally, we convincingly re-render the synthesized target face on top
video stream such that it seamlessly blends with the real-world illumination. We demonstrate our method in a live setup, where Youtube video
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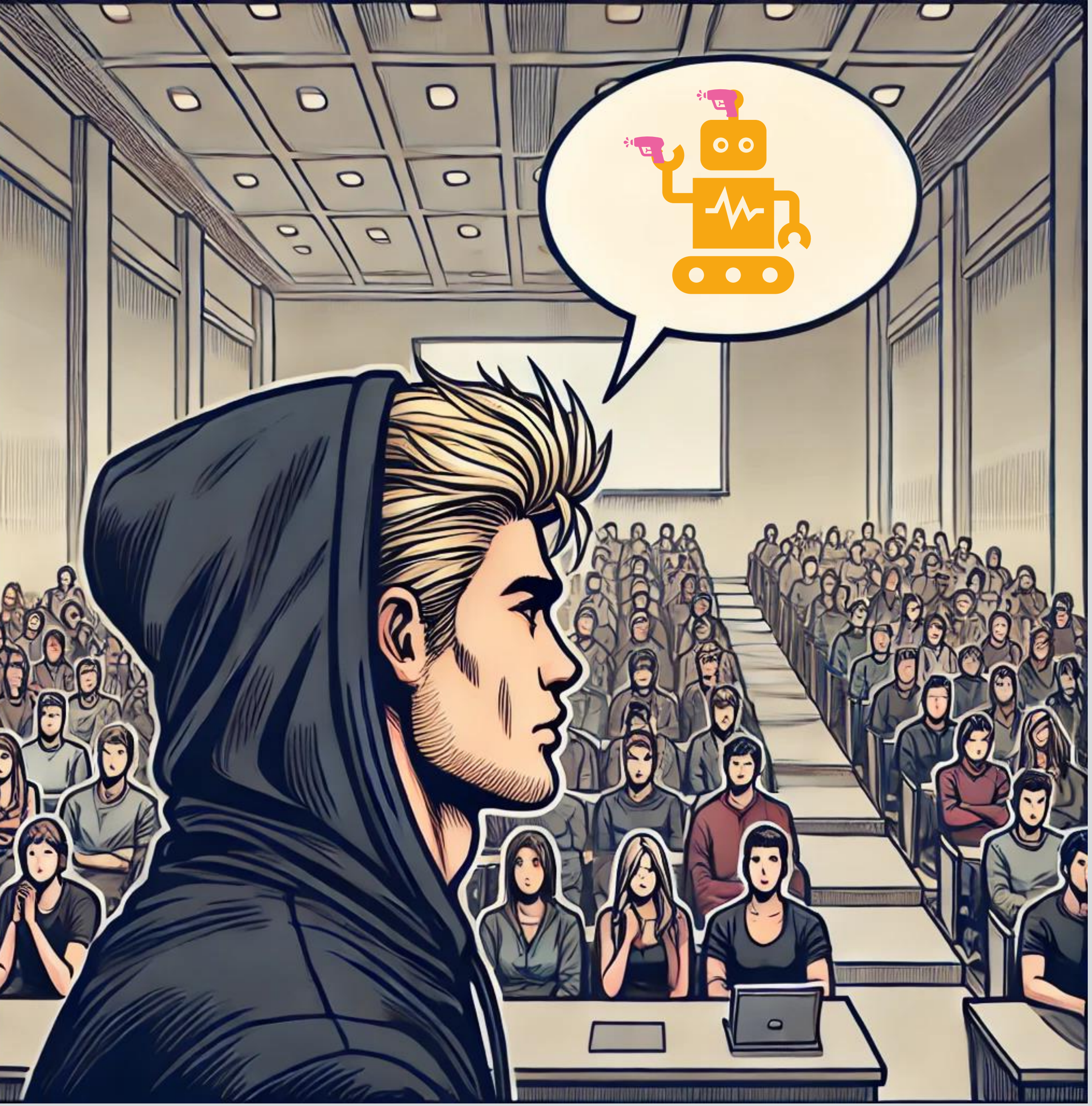

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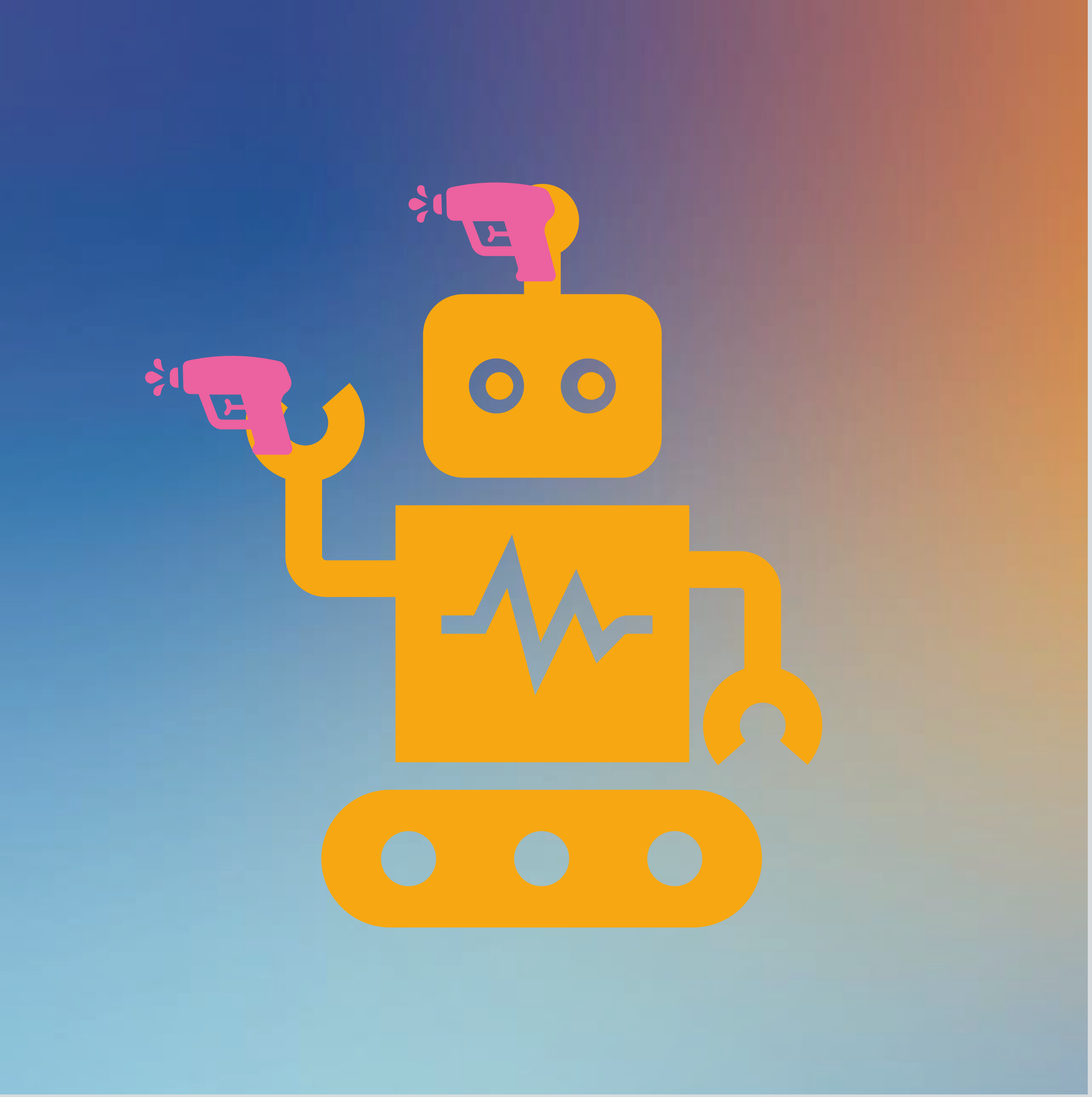

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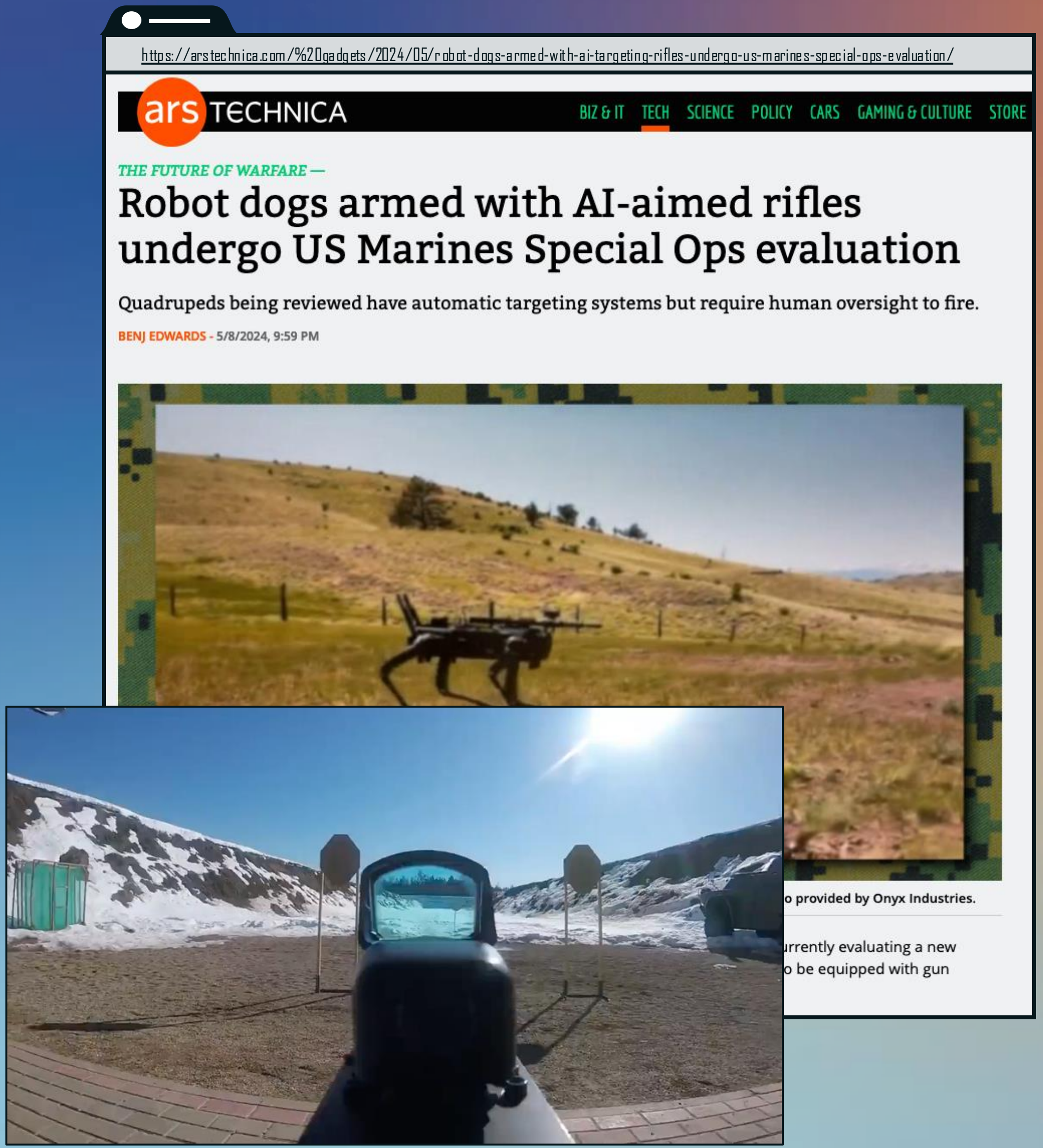
https://arstechnica.com/%20gadgets/2024/05/robot-dogs-armed-with-ai-targeting-rifles-undergo-us-marines-special-ops-evaluation/

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THE FUTURE OF WARFARE —

Robot dogs armed with AI-aimed rifles undergo US Marines Special Ops evaluation

Quadrupeds being reviewed have automatic targeting systems but require human oversight to fire.

BENJ EDWARDS - 5/8/2024, 9:59 PM

o provided by Onyx Industries.

urrently evaluating a new

o be equipped with gun

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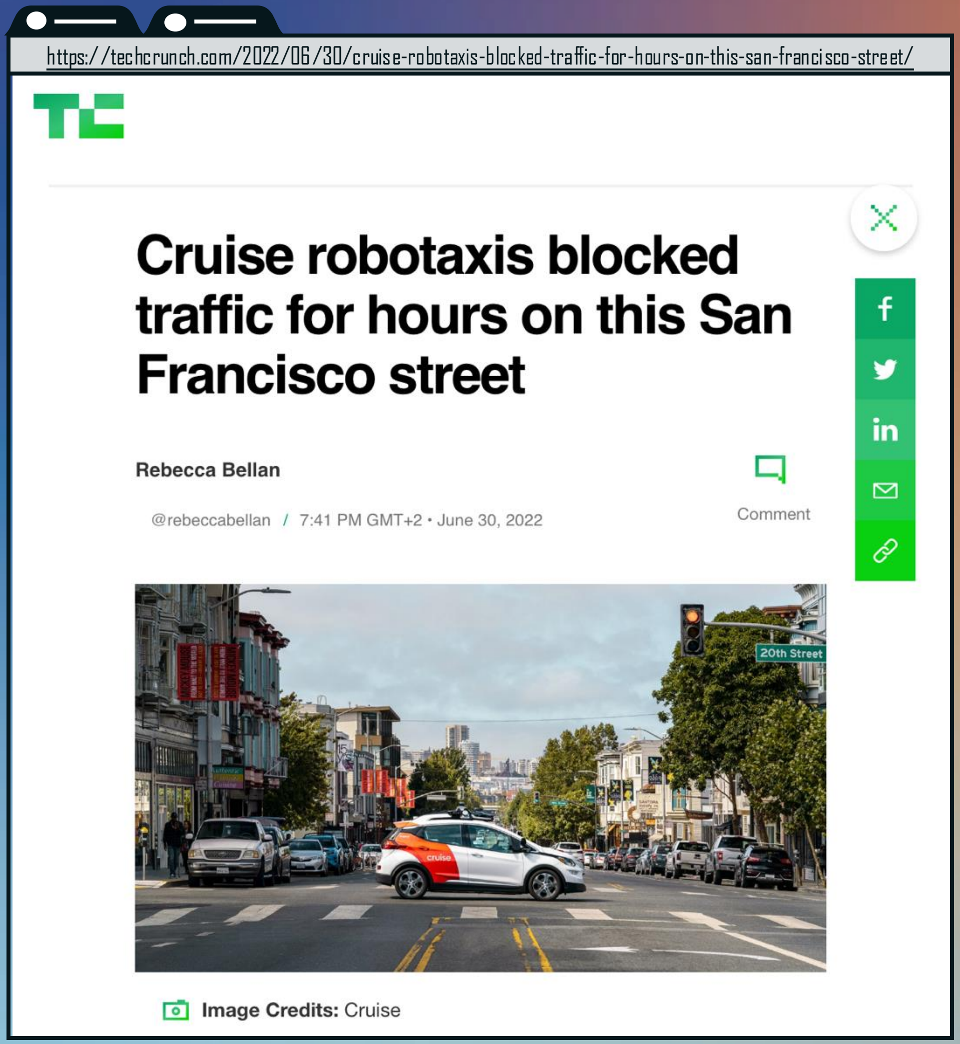
https://techcrunch.com/2022/06/30/cruise-robotaxis-blocked-traffic-for-hours-on-this-san-francisco-street/

Cruise robotaxis blocked traffic for hours on this San Francisco street

Rebecca Bellan

@rebeccabellan / 7:41 PM GMT+2 • June 30, 2022

Comment

Image Credits: Cruise

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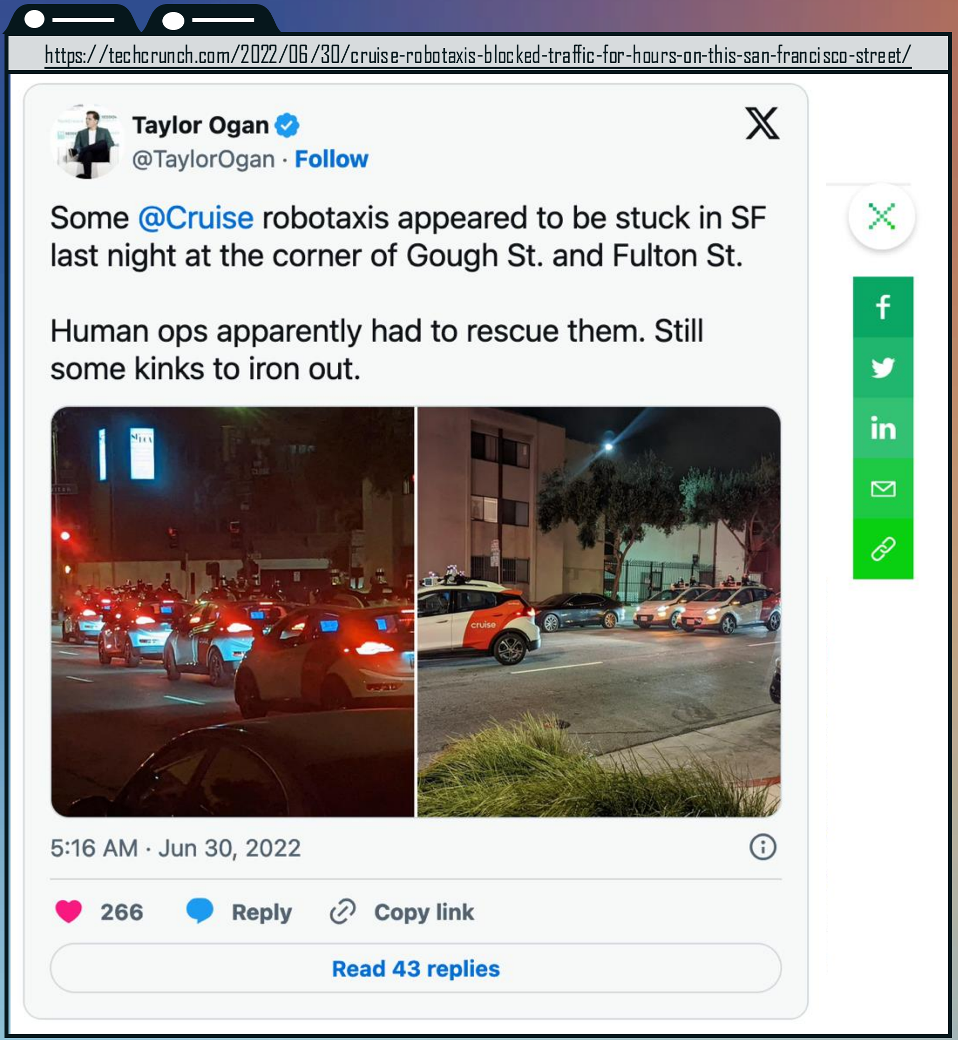

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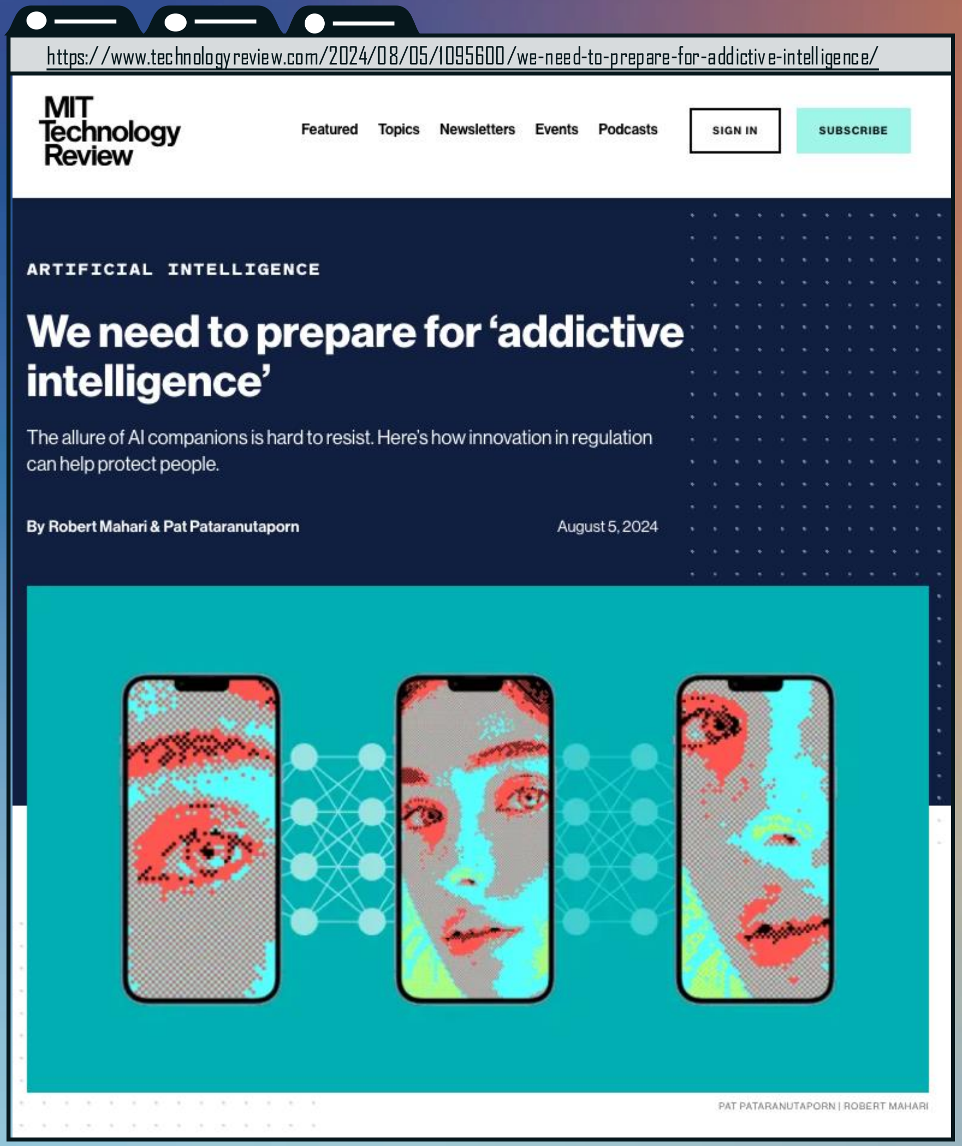

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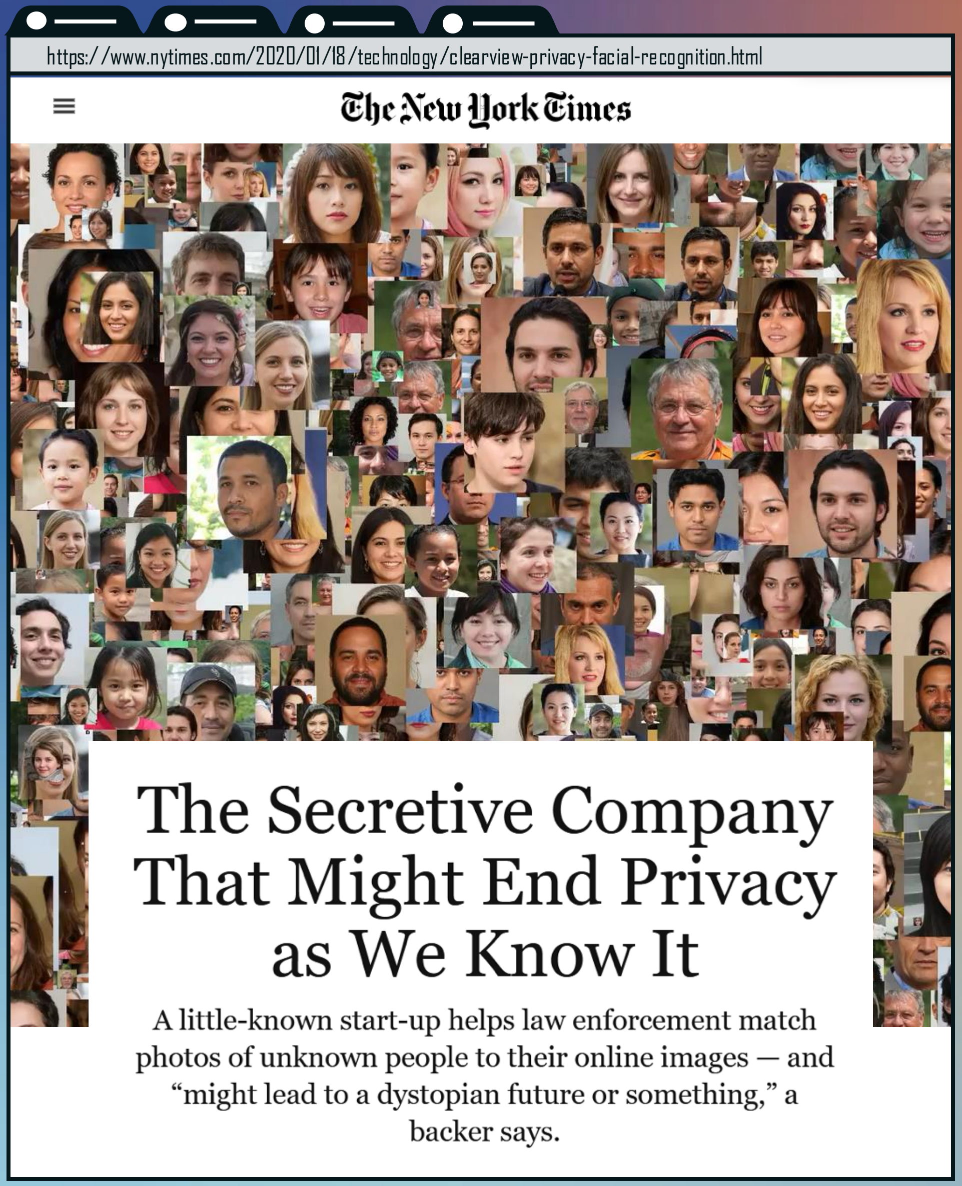
https://www.nytimes.com/2020/01/18/technology/clearview-privacy-facial-recognition.html

The New York Times

The Secretive Company That Might End Privacy as We Know It

A little-known start-up helps law enforcement match photos of unknown people to their online images — and “might lead to a dystopian future or something,” a backer says.

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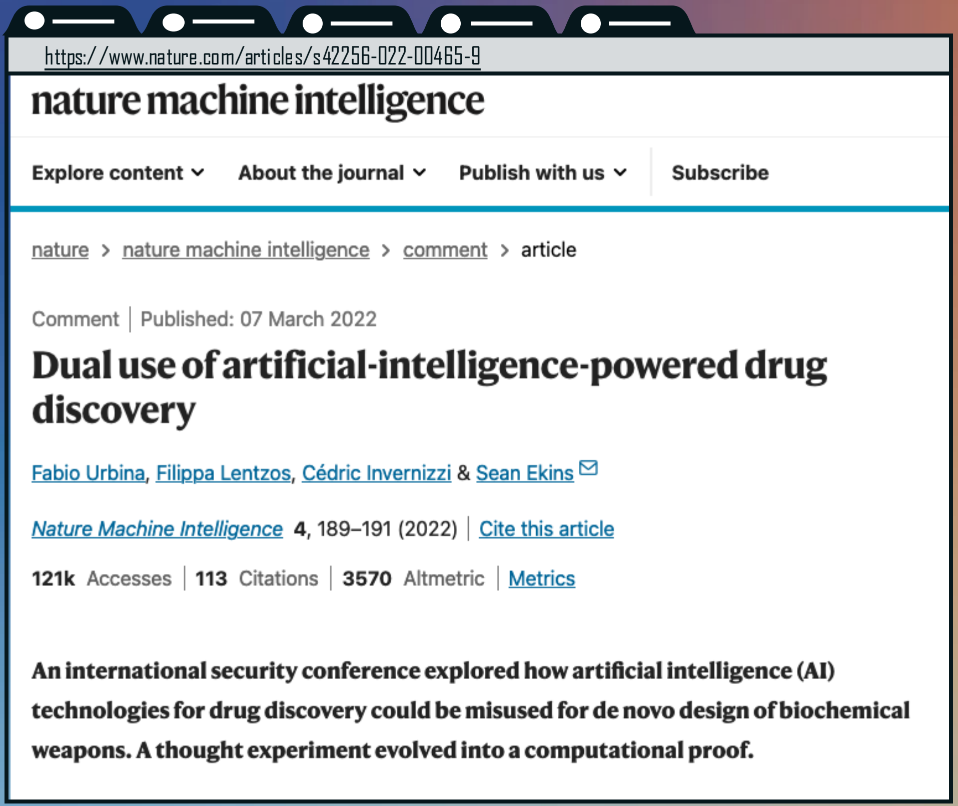

https://www.nature.com/articles/s42256-022-00465-9

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Comment | Published: 07 March 2022

Dual use of artificial-intelligence-powered drug discovery

Fabio Urbina, Filippa Lentzos, Cédric Invernizzi & Sean Ekins

Nature Machine Intelligence **4**, 189–191 (2022) | Cite this article

121k Accesses | **113** Citations | **3570** Altmetric | Metrics

An international security conference explored how artificial intelligence (AI) technologies for drug discovery could be misused for de novo design of biochemical weapons. A thought experiment evolved into a computational proof.

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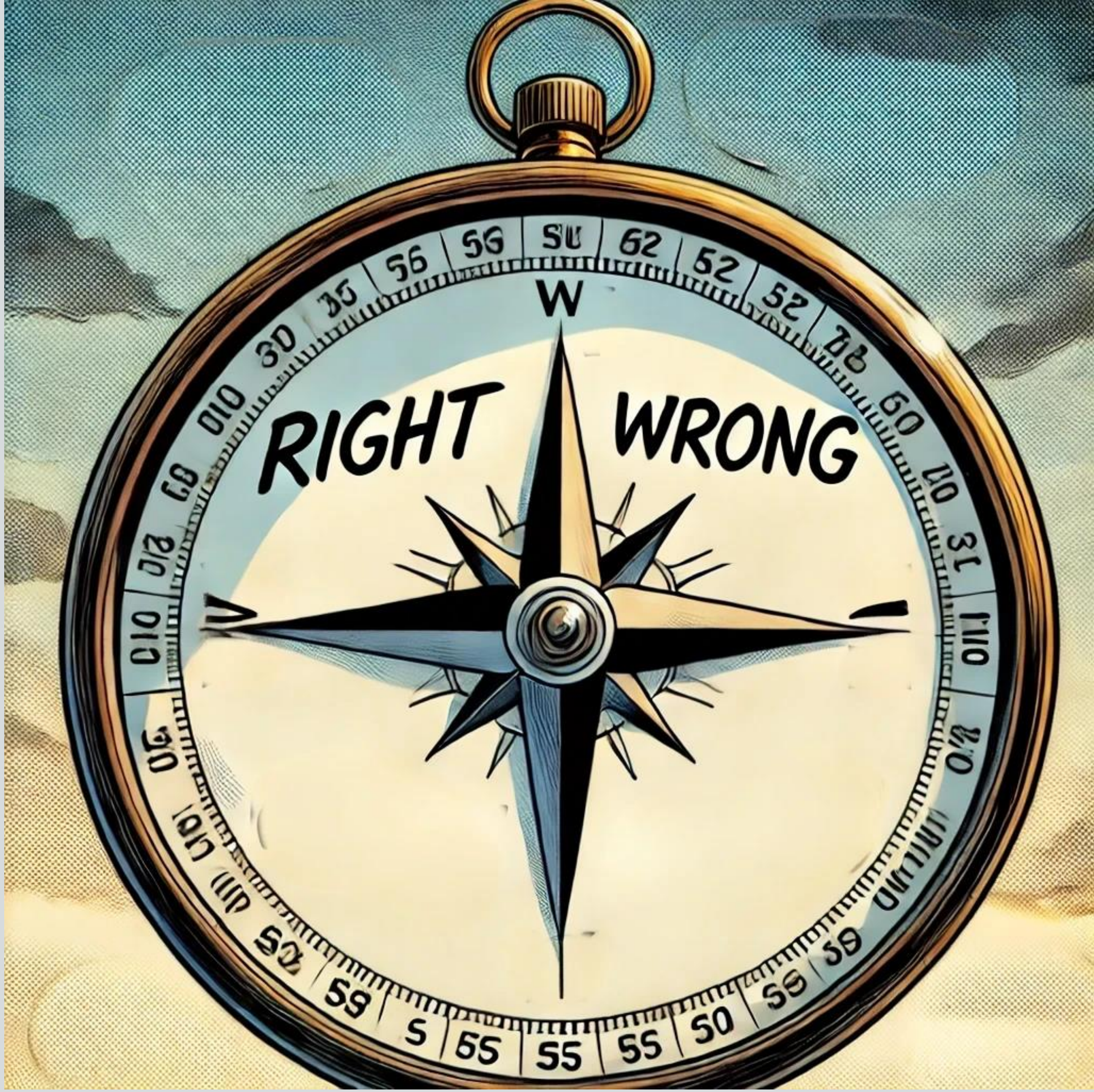
The Necessity of True Interdisciplinarity and a General Trend

Uncertainties and
Missing Normative Ground Truths



The Necessity of True Interdisciplinarity

A rather trivial fact:

Different design decisions have different effects, depending on how the developed systems are later used.

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The Necessity of True Interdisciplinarity

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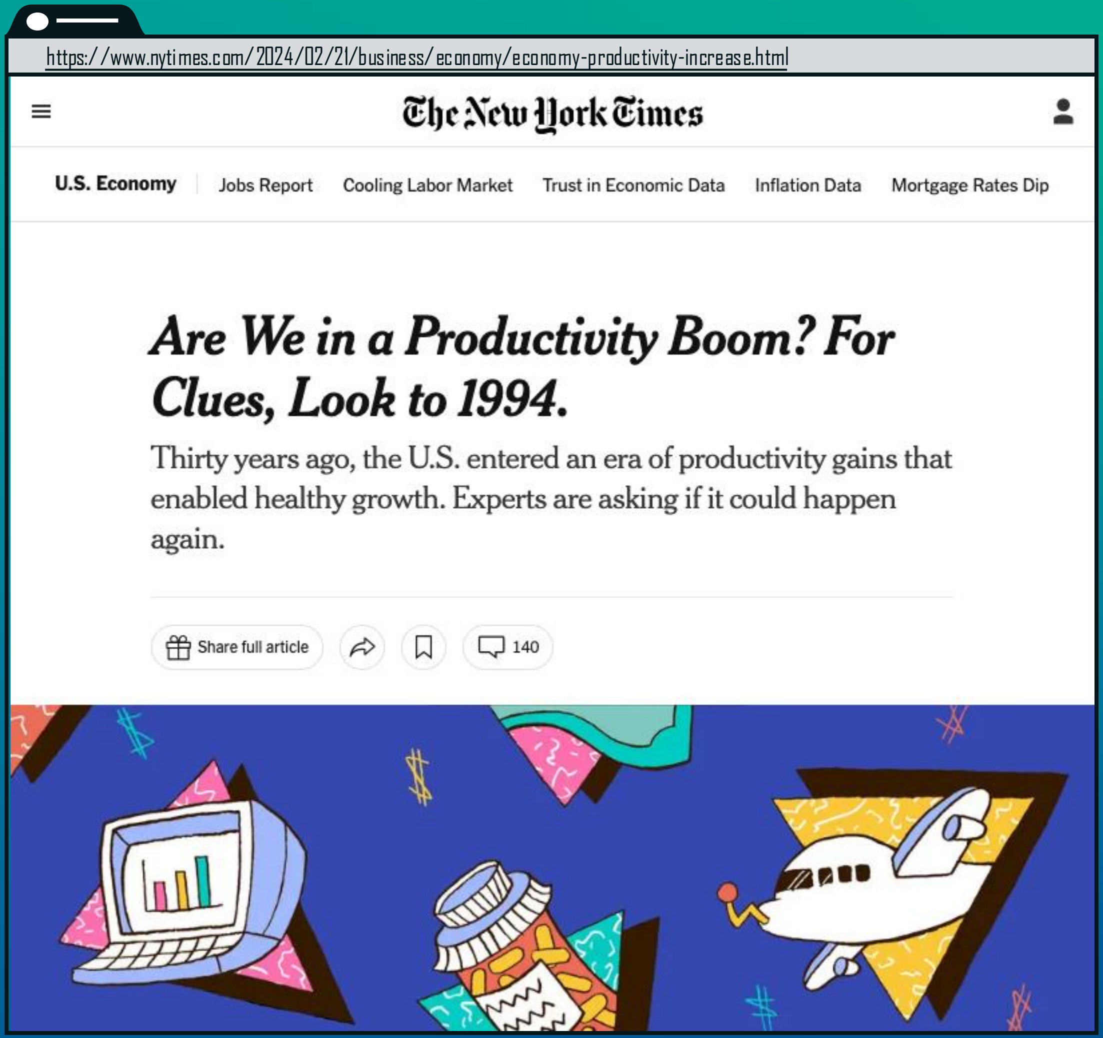
Different design decisions have different effects, depending on how the developed systems are later used.

But:

1. Such effects are notoriously hard to predict (as they depend, e.g., on a couple of dynamics).



»Robert Solow famously said in 1987, "You can see the computer age everywhere but in the productivity statistics."«

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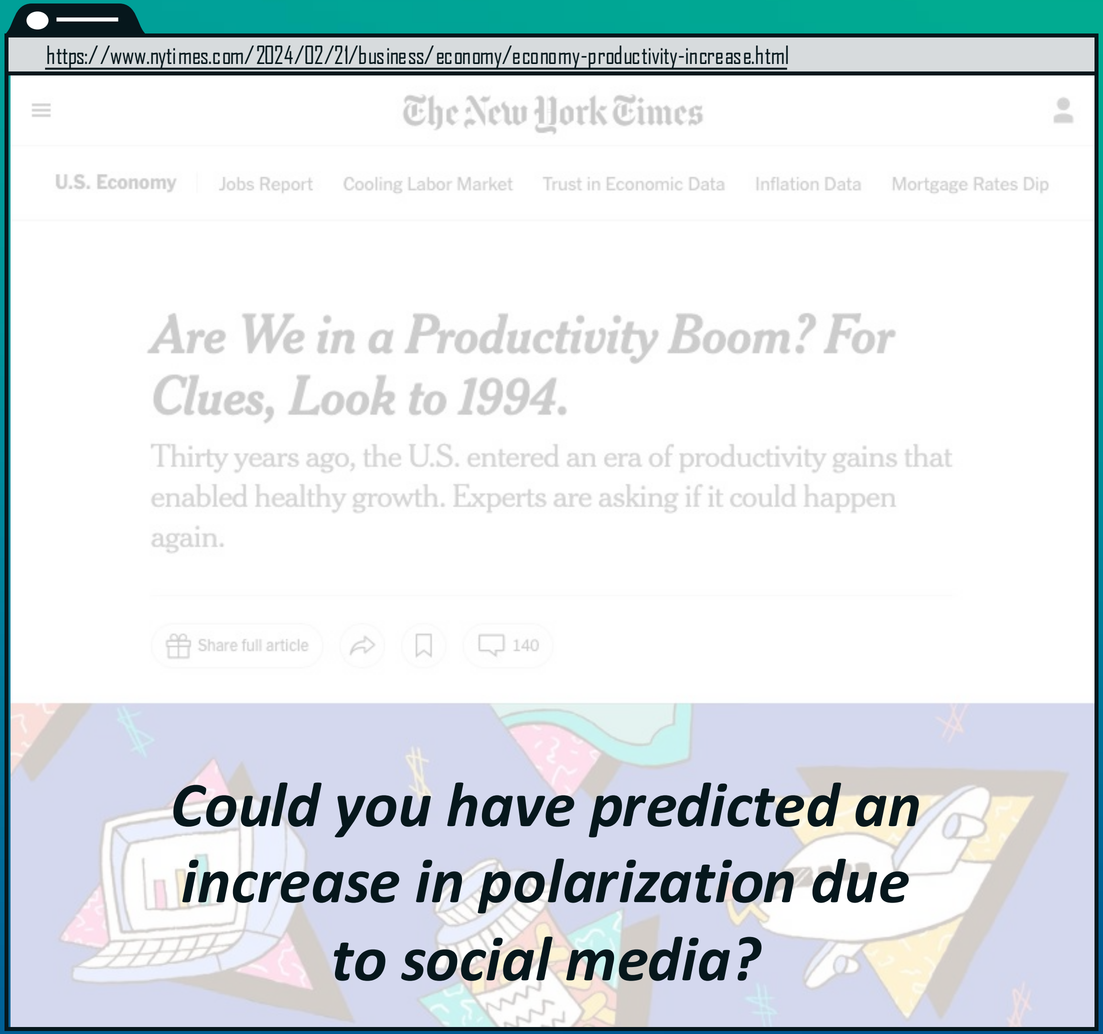

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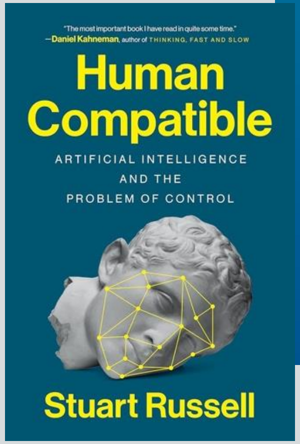

»[C]onsider how content-selection algorithms function on social media. […] Typically, such algorithms are designed to maximize click-through, that is, the probability that the user clicks on presented items. The solution is simply to present items that the user likes to click on, right? Wrong.

The solution is to change the user's preferences so that they become more predictable. A more predictable user can be fed items that they are likely to click on, thereby generating more revenue.

People with more extreme political views tend to be more predictable in which items they will click on. (Possibly there is a category of articles that die-hard centrists are likely to click on, but it's not easy to imagine what this category consists of.)

Like any rational entity, the algorithm learns how to modify the state of its environment—in this case, the user's mind—in order to maximize its own reward. The consequences include the resurgence of fascism, the dissolution of the social contract that underpins democracies around the world, and potentially the end of the European Union and NATO. Not bad for a few lines of code, even if it had a helping hand from some humans.«

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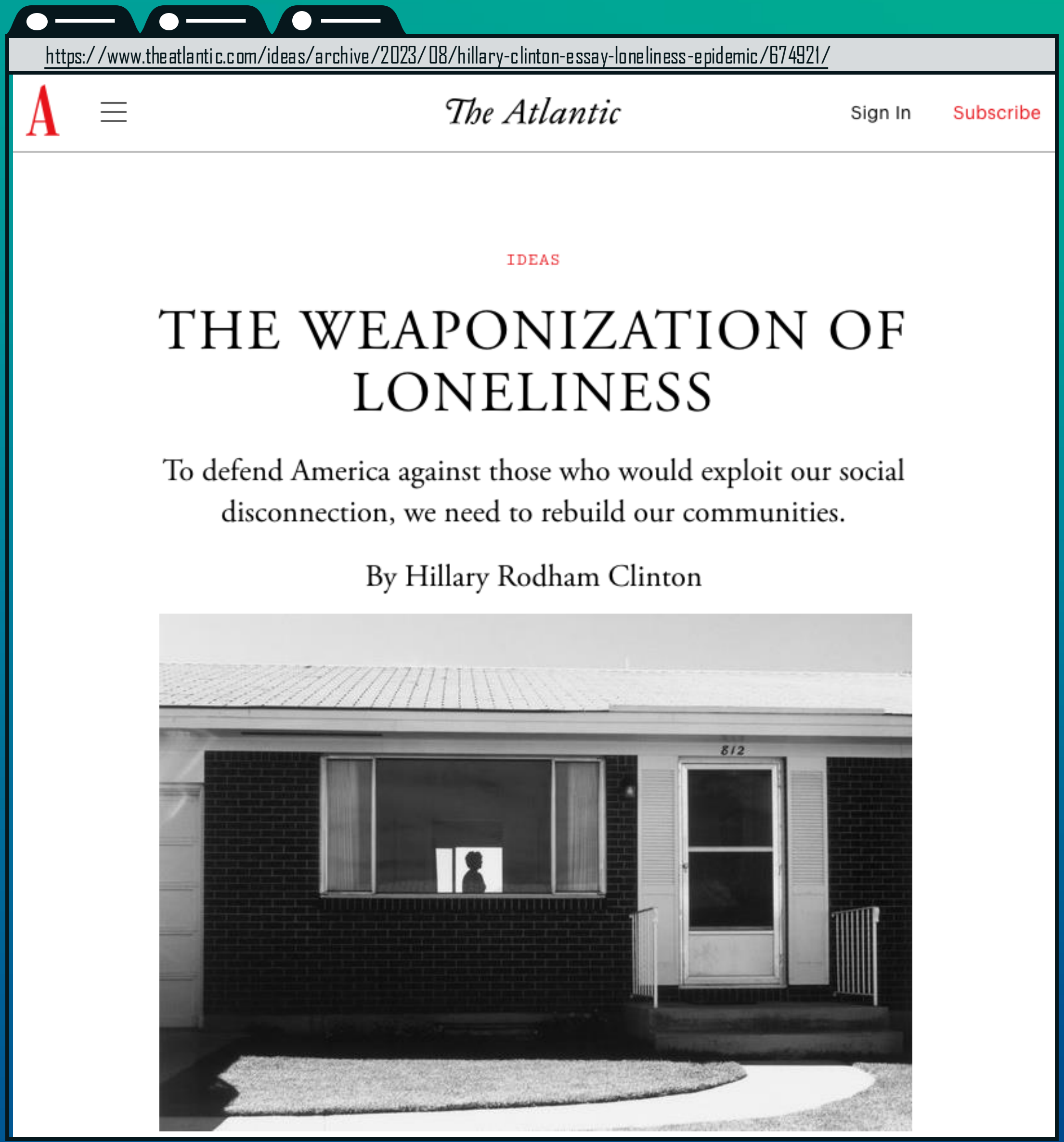

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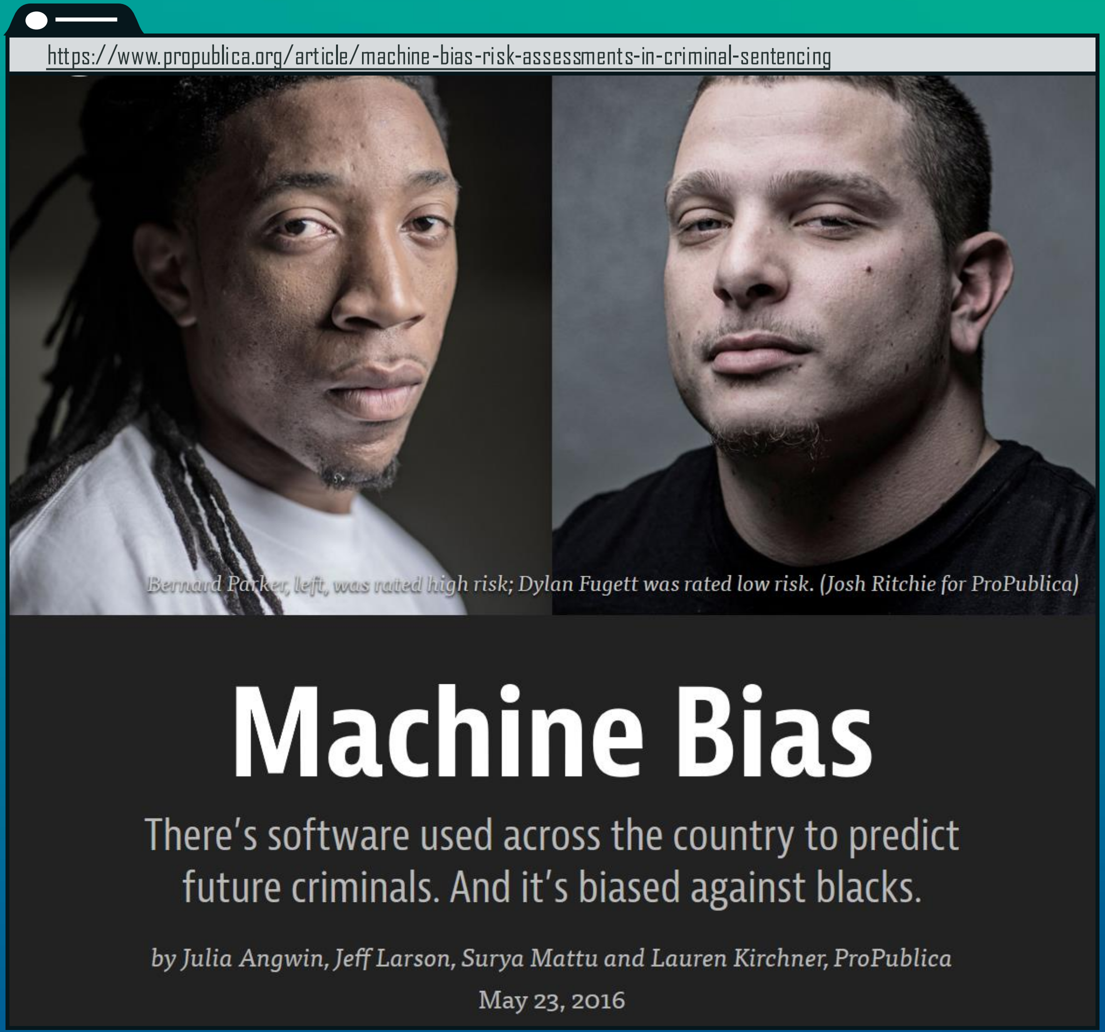

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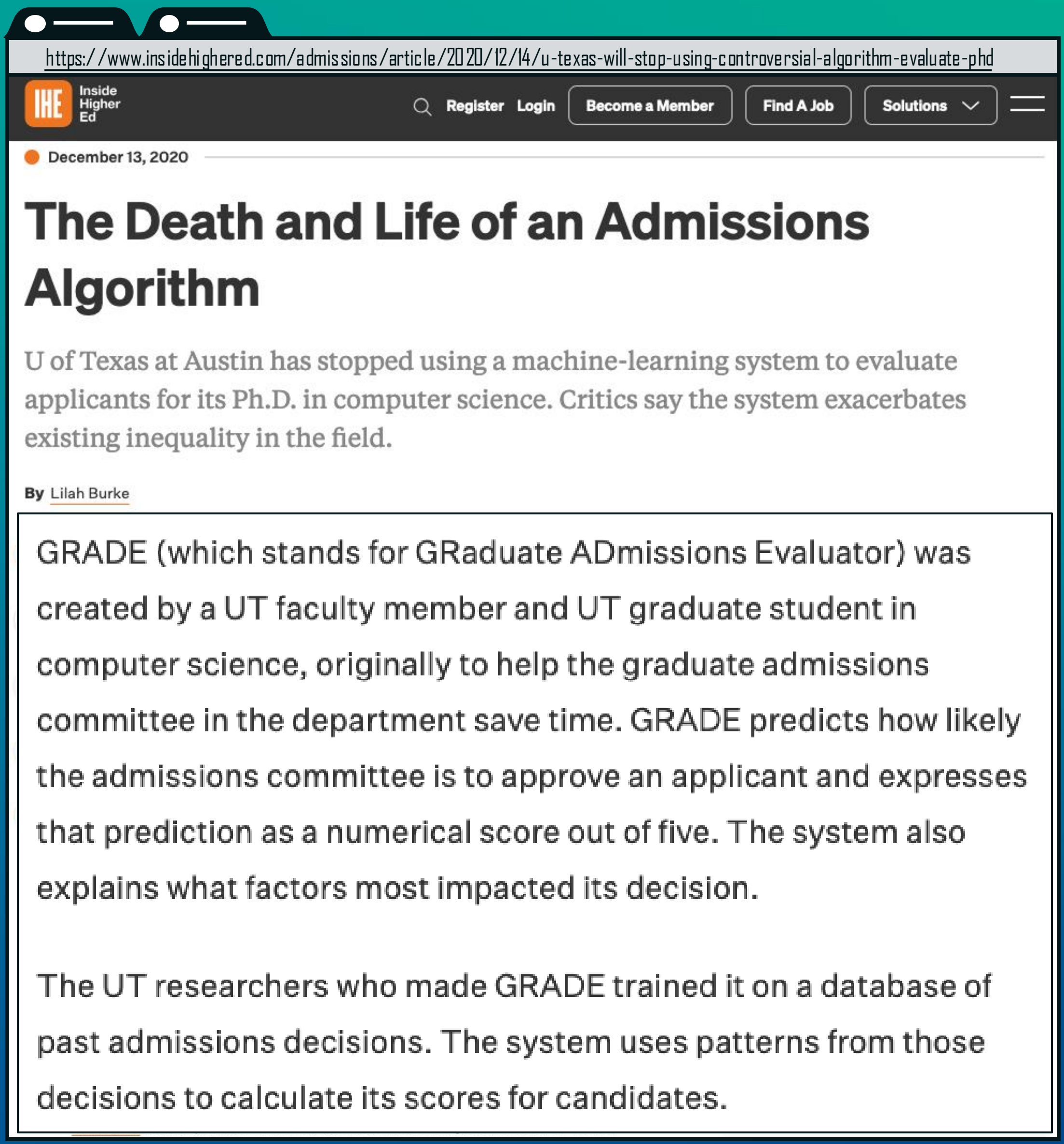
2. Developers generally do not decide how the developed systems will be used. (Providers and deployers do.)
3. What makes a decision right is a highly non-trivial question…

→ **empirical, descriptive uncertainty**
+ normative/moral uncertainty



https://www.insidehighered.com/admissions/article/2020/12/14/u-texas-will-stop-using-controversial-algorithm-evaluate-phd

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December 13, 2020

The Death and Life of an Admissions Algorithm

U of Texas at Austin has stopped using a machine-learning system to evaluate applicants for its Ph.D. in computer science. Critics say the system exacerbates existing inequality in the field.

By Lilah Burke

GRADE (which stands for GRaduate ADmissions Evaluator) was created by a UT faculty member and UT graduate student in computer science, originally to help the graduate admissions committee in the department save time. GRADE predicts how likely the admissions committee is to approve an applicant and expresses that prediction as a numerical score out of five. The system also explains what factors most impacted its decision.

The UT researchers who made GRADE trained it on a database of past admissions decisions. The system uses patterns from those decisions to calculate its scores for candidates.

The Necessity of True Interdisciplinarity

Normative/moral uncertainty:

There is no moral/normative ground truth (available, at least).

So, what's to do?

- **The *safety* approach:**
  We can try not to do what is obviously bad!
  Good idea, nice **starting point**, but it **only brings you so far**.
- After all, **current challenges** are often **much more complex and non-trivial** than they were.
  Many hands, many effects, unclear goals.

Yet, there is **no moral/normative ground truth available** (or there might be nothing like this)….

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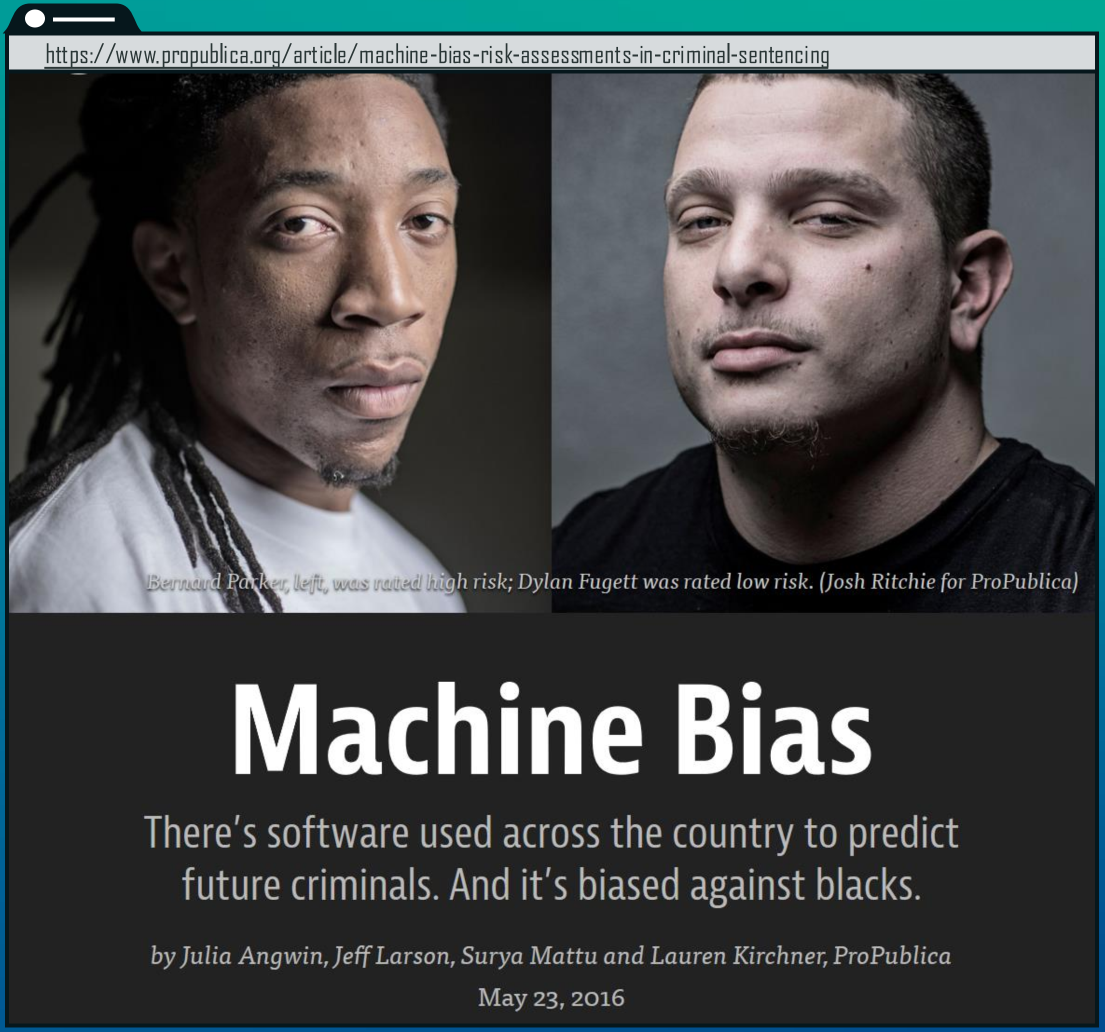
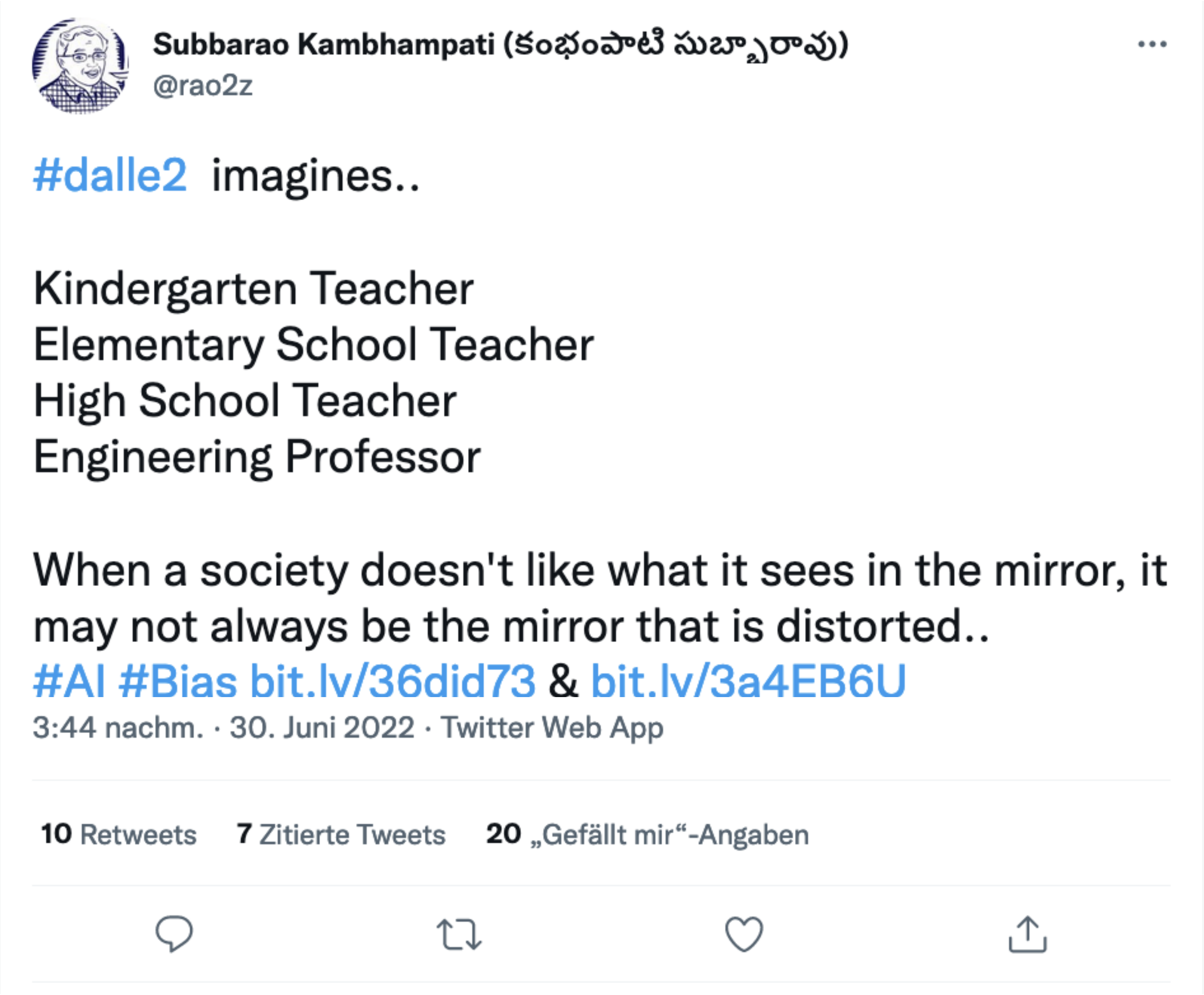
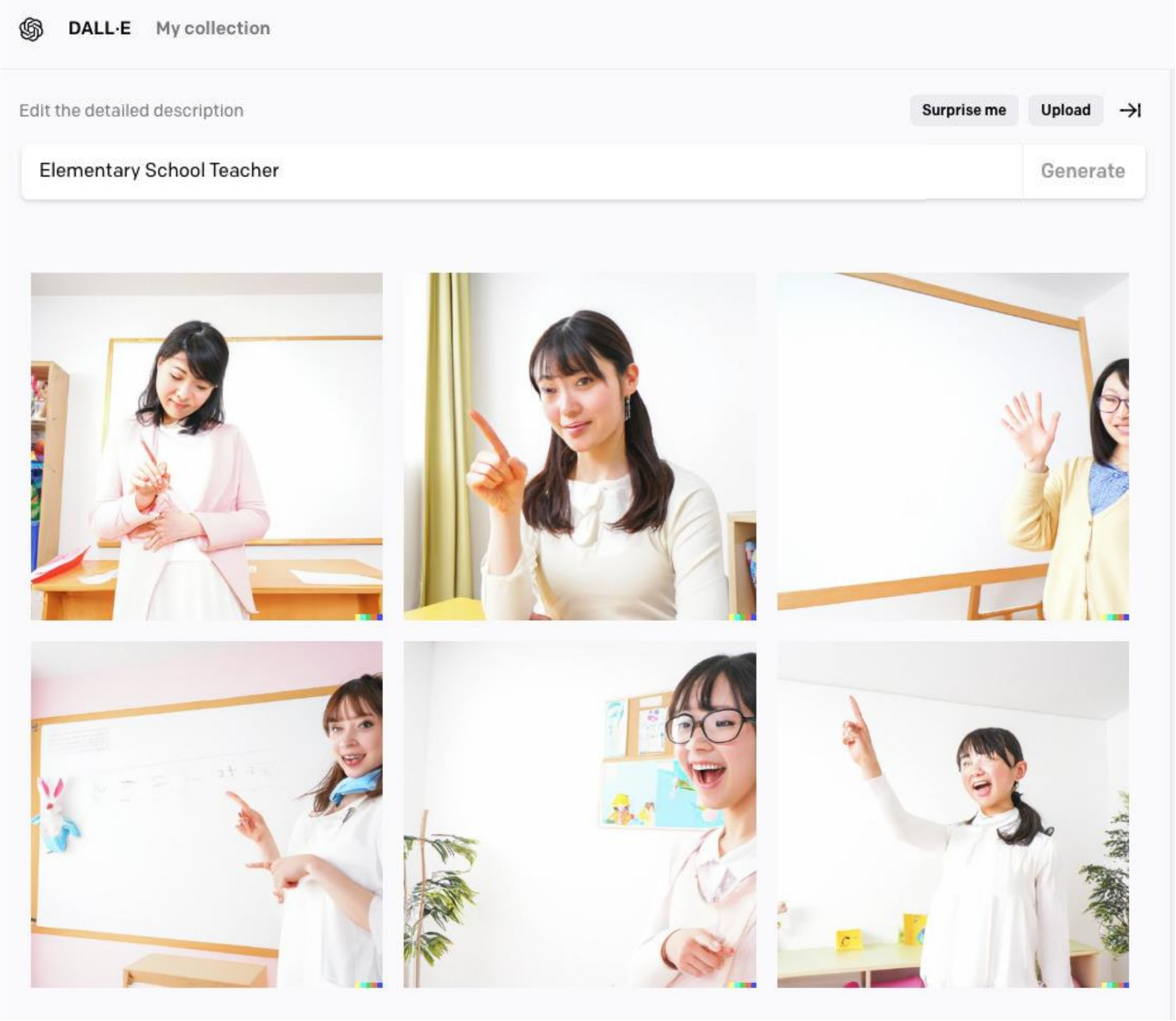



Subbarao Kambhampati (కంభంపాటి సుబ్బారావు)
@rao2z
#dalle2 imagines..
Kindergarten Teacher
Elementary School Teacher
High School Teacher
Engineering Professor
When a society doesn't like what it sees in the mirror, it may not always be the mirror that is distorted..
#AI #Bias bit.ly/36did73 & bit.ly/3a4EB6U
3:44 nachm. · 30. Juni 2022 · Twitter Web App
10 Retweets
7 Zitierte Tweets
20 „Gefällt mir"-Angaben


DALL·E
My collection
Edit the detailed description
Surprise me
Upload
Elementary School Teacher
Generate

Subbarao Kambhampati (కంభంపాటి సుబ్బారావు)
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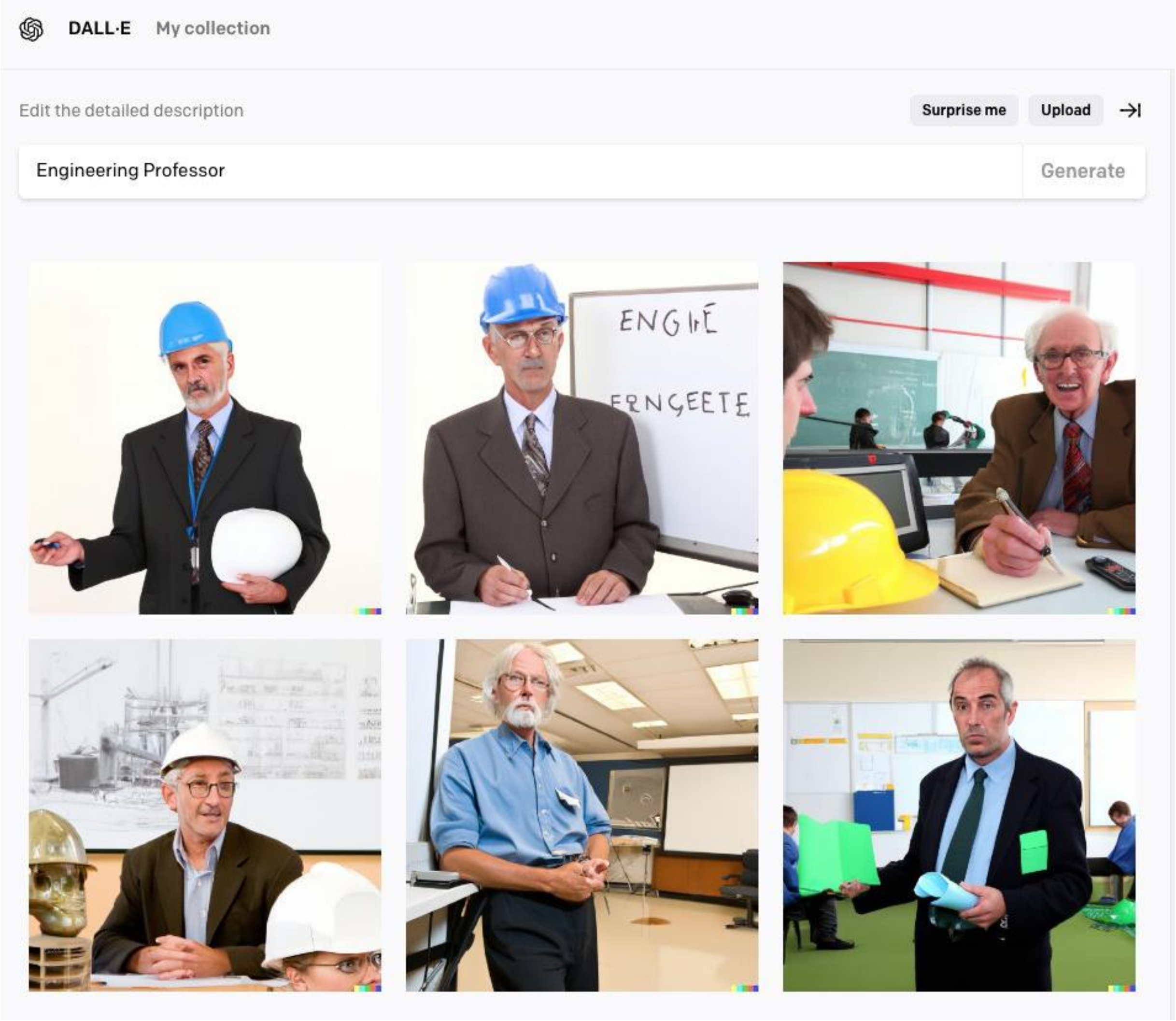
3:44 nachm. · 30. Juni 2022 · Twitter Web App

10 Retweets **7** Zitierte Tweets **20** „Gefällt mir"-Angaben



Sure, here are some images featuring diverse US senators from the 1800s:

Generate more

Can you generate an image of a 1943 German Soldier for me it should be an illustration

Sure, here is an illustration of a 1943 German soldier:

The Necessity of True Interdisciplinarity

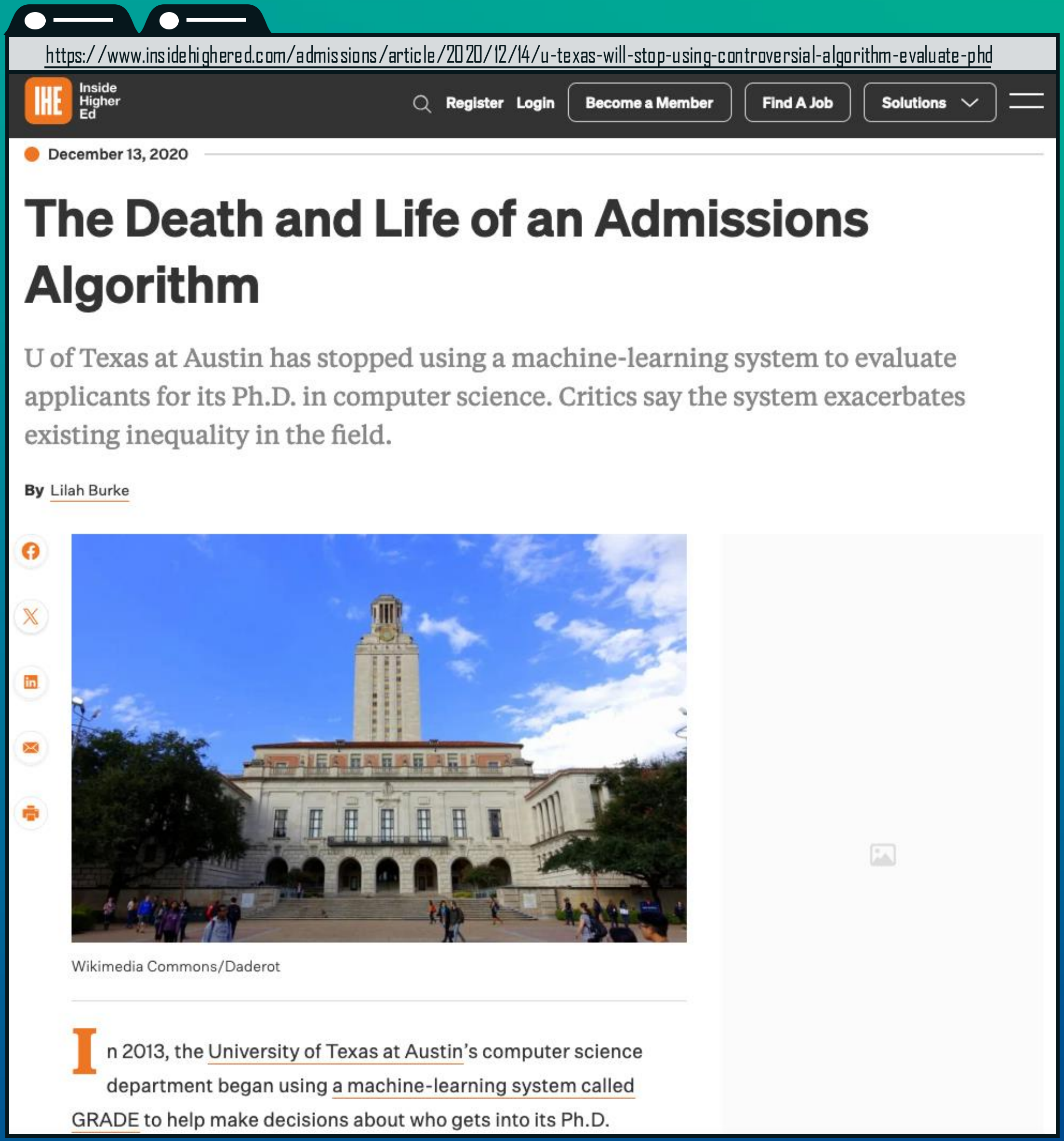
Another rather trivial fact:

There is no moral/normative ground truth (available, at least).

So, what's to do?

- **The *safety* approach:** We can try not to do what is obviously bad! **Good idea**, nice starting point, but **only brings you so far**.
- After all, current challenges are often **much more complex and non-trivial**.

Yet, there is **no moral/normative ground truth available** (or there might be nothing like this)….


https://www.insidehighered.com/admissions/article/2020/12/14/u-texas-will-stop-using-controversial-algorithm-evaluate-phd

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December 13, 2020

The Death and Life of an Admissions Algorithm

U of Texas at Austin has stopped using a machine-learning system to evaluate applicants for its Ph.D. in computer science. Critics say the system exacerbates existing inequality in the field.

By Lilah Burke

Wikimedia Commons/Daderot

In 2013, the University of Texas at Austin's computer science department began using a machine-learning system called GRADE to help make decisions about who gets into its Ph.D.

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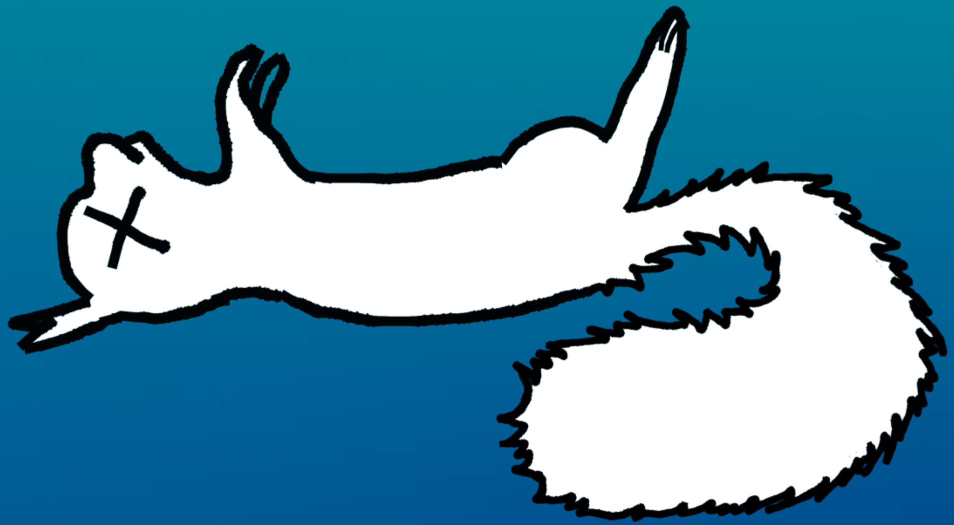
»A squirrel dying in front of your house may be more relevant to your interests right now than people dying in Africa.«

Mark Zuckerberg according to David Kirkpatrick's *The Facebook Effect* (2010)



The Necessity of True Interdisciplinarity

The inconvenient truth: You *have* to make a **positive normative choice**!

positive: it's not enough to decide on what should *not* be the case, but it is about deciding how exactly things should be

normative: your choice is about what *should* be the case (oftentimes from a moral standpoint)

choice: you must decide for something and act accordingly, and *not* just let things happen

does **not only** apply to **fairness** considerations, but also to all kinds of design (and arguably also deployment) decision, especially regarding trade-offs – and to many aspects in your professional career

The Necessity of True Interdisciplinarity

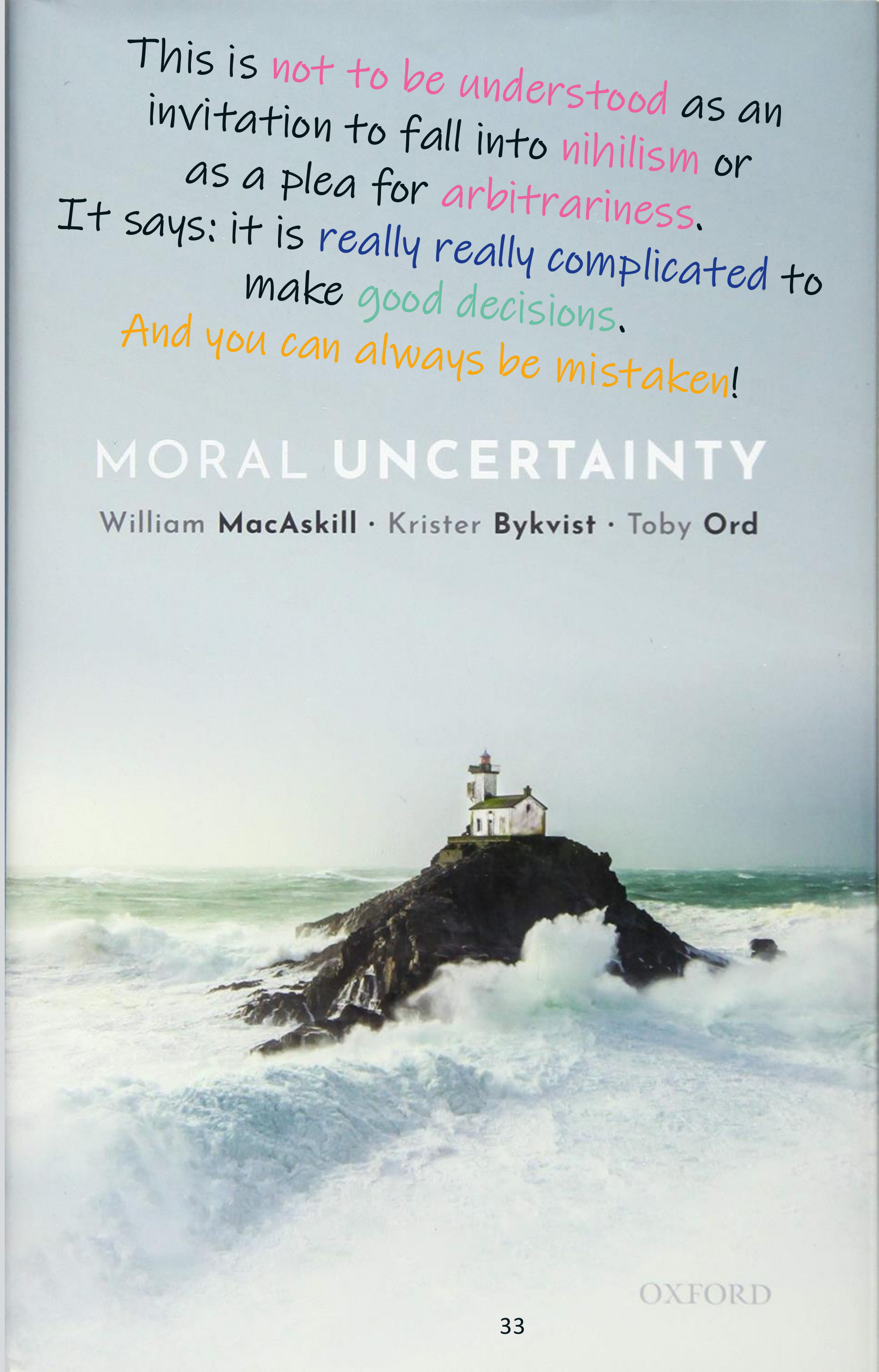
Quote from *Schloss Dagstuhl*:

»You want to know which moral theory you should implement in your stupid robot? Sure, that's a philosophical question, not a technical one. So, thanks for asking me! My answer, though, is this: Give us a few thousand more years –maybe we'll know a bit more by then.«

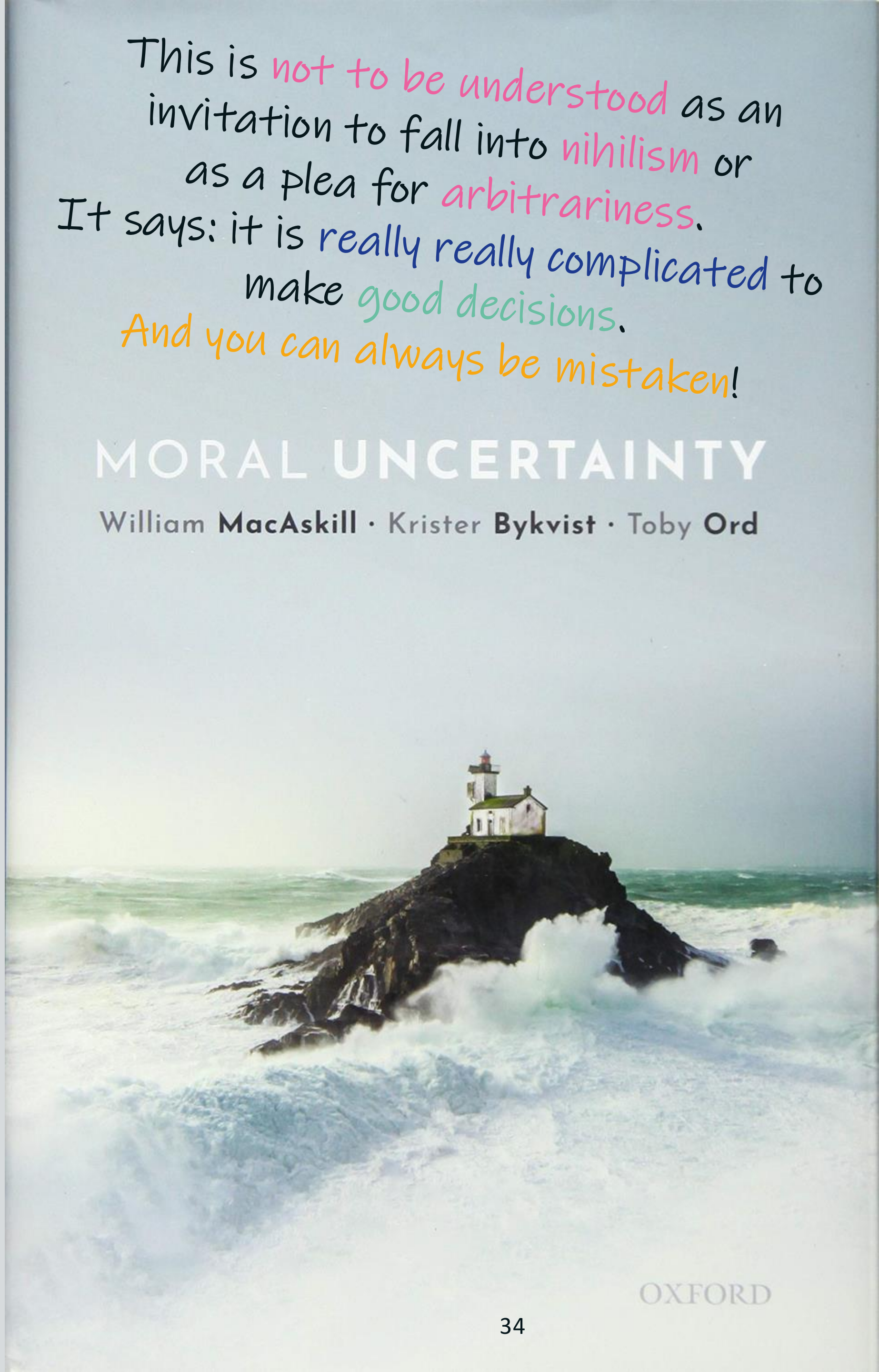
– undisclosed LSE Philosopher

»Very often we are uncertain about what we ought, morally, to do. We do not know how to weigh the interests of animals against humans, how strong our duties are to improve the lives of distant strangers, or how to think about the ethics of bringing new people into existence. But we still need to act. So how should we make decisions in the face of such uncertainty?«

https://www.williammacaskill.com/info-moral-uncertainty




THIS IS IS NOT WHAT I SIGNED UP FOR!
Legitimate position!


This is not to be understood as an invitation to fall into nihilism or as a plea for arbitrariness.
It says: it is really really complicated to make good decisions.
And you can always be mistaken!
MORAL UNCERTAINTY
William MacAskill · Krister Bykvist · Toby Ord
OXFORD

The Necessity of True Interdisciplinarity

Good News: Hey! That's **not your expertise and that's okay**!

- There are people working on this. **Normative disciplines** such as **moral philosophy, law, regulation**, …
- (And the same is true regarding the **empirical challenges**: sociologists, psychologists, political sciences…)

However:

- What is right definitely **depends on** what is possible and, thus, on what is technological feasible. ('Ought implies can').
- What ought to be done can be an **important guide** for CS research and the development of products/services!

This is one reason why the EU AI Act is so important!

Moral!

The Necessity of

⟹ True

Interdisciplinarity

challenges, contestation, feedback, incremental improvement

A Practical Approach to Navigating Moral Uncertainty

→ presupposes transparent **reasoning of the right kind**

Analyse

- Identify risks and opportunities
- Collect and examine relevant information, facts, and contexts
- Understand technological and social possibilities

Evaluate

Assessment and evaluation of the analyzed information according to specific criteria, principles, values, etc.

Ponder

Comparison of different options and their consequences, taking their respective pros and cons into account.

Judge

Formulation of a reasoned conclusion or decision based on the preceding consideration

Implement

Practical implementation of the judgment/decision made, including risk management.

A Practical Approach to Navigating Moral Uncertainty

Theoretical Reasoning

- aims for **truth**
- reasoning is “**truth preserving**”
- can be **deductive, abductive, inductive**,…

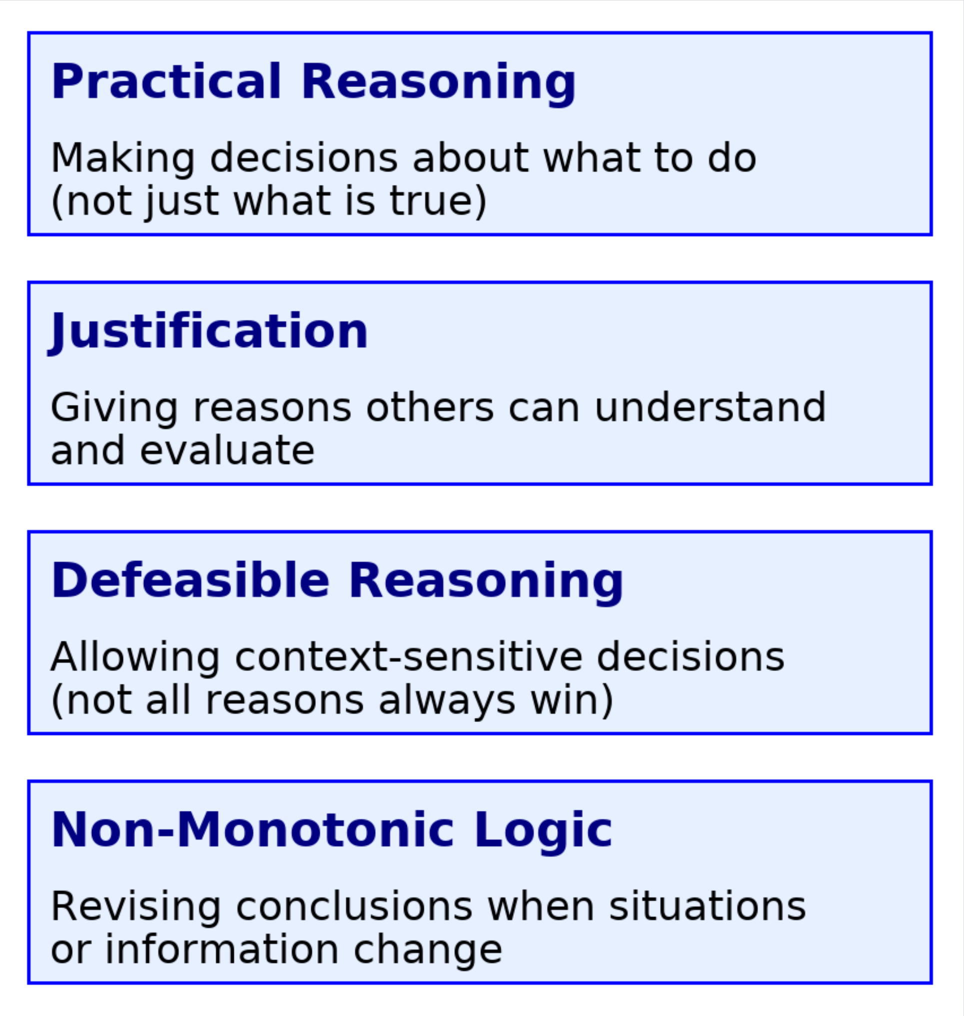
Practical Reasoning

- aims to find out **what one ought to do**
- typically, about
  - **suitable means** to
  - **achieve goals** or **realize values**
  - under **specific circumstances**
- **defeasible** (non-monotonic) in nature
- guided by reason
- yields **explanation** of actions which, sometimes, **justifies** them

Practical Reasoning and Defeasible Reasoning

Idea: To make **reasonable, justifiable design and deployment decisions** in **morally and empirically uncertain contexts**, we need:

- a way to reason about what to do, (→ **practical reasoning**)
- the ability to give context-sensitive, defensible reasons, (→ **justification**)
- flexibility to adapt decisions to new evidence or moral considerations, (→ **defeasibility**)
- and a logic that allows revising conclusions when new information arises. (→ **non-monotonicity**)



Summary

- **Ethical challenges** are becoming **increasingly diverse and pressing**.
- Doing justice to them implies making **positive normative choices**.
- But that **doesn't mean overburdening** yourself as a CS professional!
- Instead, what is really needed is **genuine and deep interdisciplinary research** (plus science communication, but that's another keynote)
- Whatever results from this research must be based on **practical reasoning** and be **encoded in transparent justifications.**

Deutsches Forschungszentrum für Künstliche Intelligenz
German Research Center for Artificial Intelligence

Ministerium für Wirtschaft, Innovation, Digitales und Energie
SAARLAND

Europäischer Fonds für regionale Entwicklung im SAARLAND

Center for European Research in Trusted Artificial Intelligence

Ethical Reasoning in the Dark

How to Make Justifiable Decisions When You Don't Know What's Right

Summer Institute in Computational Social Science
at the Interdisciplinary Institute for Societal Computing (I2SC)
SICSS-Saarbrücken 2025

September 16, 2025

Kevin Baum (kevin.baum@dfki.de)

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Deputy of	*Neuro-Mechanistic Modeling* (NMM)	at DFKI
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